

Linear algebra [BCSE SC 404B] - ③

Module - 01

Vector space

- 1) Introduction to vector space, subspace,
- * linear combination, linear span, row space & column space of a matrix.
- * linear dependency and independency. $\rightarrow$ (determinant = 0)
- * Basis and dimension.

vector space :—

- Group : ~~non~~ non-empty set with one binary operation $*$ is called "group". It satisfies the following conditions @ properties -
 - i) Closure property : $\forall a, b \in G, a * b \in G$
 - ii) Associative property
 $\forall a, b, c \in G$
 $a * (b * c) = (a * b) * c$
 - iii) Identity property : $\forall a \in G \Rightarrow e * a = a \in G$
 - iv) Inverse property : $a \in G, a^{-1} \in G \Rightarrow a * a^{-1} \in G$
 - v) commutative property : $\forall a, b \in G \Rightarrow a * b = b * a$

Imp ③

* Define vector space.

$\rightarrow$ Let F is a field and V is a non empty set with two binary operations - Addition and - Scalar multiplication

such that $u, v \in V$ there exists unique element $u + v \in V$. And for each $\alpha \in F, u \in V$ there exist unique element $\alpha * u \in V$ and it satisfies the following conditions.

i) $(V, +)$ is group (Addition property)

ii) $c(\alpha + \beta) = c\alpha + c\beta$ (Scalar multiplication)
 $\forall \alpha, \beta \in V$

iii) $(c_1 + c_2)\alpha = c_1\alpha + c_2\alpha$

iv) $(c_1 \cdot c_2)\alpha = c_1(c_2\alpha)$

v) $1 \times \alpha = \alpha$ (Identity property)

For ex:- i) Every field is a vector space itself.

ii) Real numbers over a real line is a vector space.

Subspace :— A non-empty subset W of a vector space V is said to be subspace, if W is also a vector space with addition and scalar multiplication property.

Ex:- $W_1 = \{a, 0, 0\}, a \in \mathbb{R}$

$W_2 = \{a, b, 0\}, a, b \in \mathbb{R}$

Here, W_1 is a subspace of W_2 .

Necessary & sufficient conditions for subspace:

i) Theorem 1 : W is a subspace of vector space V , if and only if (IFF).

• W is a non-empty set.

• W is closure addition. $W_1, W_2 \in W, W_1 + W_2 \in W$.

• W is a scalar multiplication.

$\forall a \in F, w \in W, ax \in W$.

Problems:-

1) $V = \{x, y, z \in \mathbb{R}^3 : 2x + 3y^2 - 4z^2 = 0\}$ is a vector space or not?

$\Rightarrow$ For ex. $\begin{matrix} (2, 0, 1) \\ x, y, z \end{matrix}$ and $\begin{matrix} (8, 0, 2) \\ x, y, z \end{matrix} \in V$

$$4 + 0 - 4 = 0$$

$$16 + 0 - 16 = 0$$

$$(2, 0, 1) + (8, 0, 2) = (10, 0, 3)$$

x, y, z

$$20 + 0 - 16 \neq 0$$

$\therefore$ The given polynomial is not a vector space.

Linearly dependency:- A set $\{a_1, a_2, a_3, \dots, a_n\}$ of a vector space V is said to be linearly ~~dependency~~ if there exists a scalar $c_1, c_2, c_3, \dots, c_n$ such that $a_1c_1 + a_2c_2 + a_3c_3 + \dots + a_nc_n = 0$

* $\boxed{\rho(A) < \text{Number of unknowns.}}$

* determinant, $\det(S) = 0 \rightarrow$ (row must be equal to column)

Linearly independence:- When $\det(S) \neq 0$.

* $\boxed{\rho(A) = \text{Number of unknowns.}}$

• A set $\{a_1, a_2, a_3, \dots, a_n\}$ is said to be linearly independent in vector space V if there exists $a_1c_1 + a_2c_2 + a_3c_3 + \dots + a_nc_n = 0$ when $c_1 = 0, c_2 = 0, c_3 = 0, \dots, c_n = 0$.

Problem

Identify whether the following set of vectors are linearly independent.

$\{(2, 2, 1), (-4, 6, 5), (1, 0, 0)\}$

Soln:- $V_1 = (2, 2, 1)$, $V_2 = (-4, 6, 5)$, $V_3 = (1, 0, 0)$

$$Ax = 0$$

$$x(2, 2, 1) + y(-4, 6, 5) + z(1, 0, 0) = (0, 0, 0)$$

$$2x - 4y + z = 0$$

$$2x + 6y + 0z = 0$$

$$x + 5y + 0z = 0$$

$$A = \begin{bmatrix} 2 & -4 & 1 \\ 2 & 6 & 0 \\ 1 & 5 & 0 \end{bmatrix}$$

$$\therefore \rho(A) = 3$$

$$R_3 \leftrightarrow R_1$$

Number of unknowns = 3

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 6 & 0 \\ 2 & -4 & 1 \end{bmatrix}$$

$$\rho(A) = 3 = \text{Number of unknowns}$$

Hence, the given vectors are linear independent.

$$R_2 = R_2 - 2R_1$$

$$R_3 = R_3 - 2R_1$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -4 & 0 \\ 0 & -14 & 1 \end{bmatrix}$$

$$R_3 = 4R_3 - 14R_2$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Q) Show that the set $W = \{(x, y, z) : x - 4y + 7z = 0\}$ of a vector space is a subspace of $V_3(R)$

Soln:- $u = (x_1, y_1, z_1)$ and $v_2 = (x_2, y_2, z_2) \in W$

$$x_1 - 4y_1 + 7z_1 = 0 \quad \text{and} \quad x_2 - 4y_2 + 7z_2 = 0$$

For any two constants c_1 & c_2
 consider $c_1 u + c_2 v = c_1(x_1, y_1, z_1) + c_2(x_2, y_2, z_2)$
 $= (c_1 x_1 + c_2 x_2, c_1 y_1 + c_2 y_2, c_1 z_1 + c_2 z_2)$
 $= (c_1 x_1 + c_2 x_2, c_1 y_1 + c_2 y_2, c_1 z_1 + c_2 z_2)$
 $\Rightarrow (c_1 x_1 + c_2 x_2) + 4(c_1 y_1 + c_2 y_2) + 7(c_1 z_1 + c_2 z_2)$

$\Rightarrow c_1 x_1 + c_2 x_2 + 4c_1 y_1 + 4c_2 y_2 + 7c_1 z_1 + 7c_2 z_2 = 0$

$\Rightarrow c_1 x_1 + 4c_1 y_1 + 7c_1 z_1 + c_2 x_2 + 4c_2 y_2 + 7c_2 z_2 = 0$

$c_1 [x_1 + 4y_1 + 7z_1] + c_2 [x_2 + 4y_2 + 7z_2] = 0$

$c_1 [0] + c_2 [0] = 0 \Rightarrow 0 = 0$

RHS = LHS

$\therefore$ The given vector space is subspace.

3) Show that the $W = \{(x, y, z) \mid (3x + 5y + z = 0)\}$ of a vector space is a subspace.

Solⁿ:- let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$

$3x_1 + 5y_1 + z_1 = 0$ and $3x_2 + 5y_2 + z_2 = 0$

For any two constants c_1 & $c_2 \in \mathbb{R}$

consider $c_1 u + c_2 v = c_1(x_1, y_1, z_1) + c_2(x_2, y_2, z_2)$

$= (c_1 x_1, c_1 y_1, c_1 z_1) + (c_2 x_2, c_2 y_2, c_2 z_2)$

$= (c_1 x_1 + c_2 x_2, c_1 y_1 + c_2 y_2, c_1 z_1 + c_2 z_2)$

$\Rightarrow 3(c_1 x_1 + c_2 x_2) + 5(c_1 y_1 + c_2 y_2) + (c_1 z_1 + c_2 z_2) = 0$

$3c_1 x_1 + 5c_1 y_1 + c_1 z_1 + 3c_2 x_2 + 5c_2 y_2 + c_2 z_2 = 0$

$c_1 [3x_1 + 5y_1 + z_1] + c_2 [3x_2 + 5y_2 + z_2] = 0$

$c_1 [0] + c_2 [0] = 0$

$0 + 0 = 0$

LHS = RHS

$\therefore$ The given vector space is subspace

4) Identify whether the following set of vectors are $V_1 = (1, 2, -3, 4)$, $V_2 = (3, -1, 2, 1)$, $V_3 = (1, -5, 8, -7)$ are linearly dependent in $V_4(R)$

Soln:- $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & -1 & -5 \\ -3 & 2 & 8 \\ 4 & 1 & -7 \end{bmatrix}$

It is not square matrix.

$$= \begin{bmatrix} 1 & 3 & 1 \\ 0 & -7 & -7 \\ 0 & 11 & 11 \\ 0 & -11 & -11 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1 \quad -5 - 2(1) = -5-2$$

$$R_3 = R_3 + 3R_1$$

$$R_4 = R_4 - 4R_1$$

$$4 - 4(1)$$

$$1 - 4(3)$$

$$1 - 12$$

$$-7 - 4(1)$$

$$-7 - 4$$

Write it in equation

$$R_4 = R_4 + R_3, \quad R_2 \times \left(-\frac{1}{7}\right)$$

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 11 & 11 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 = R_3 - 11R_2$$

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \rho(A) = 2$$

No. of unknowns = 4

Hence, $\rho(A) < \text{No. of unknowns}$

$\therefore$ The given vectors are linearly dependent.

5) Test whether the following set of vectors linearly dependent or independent. $(1, 1, 1, 1)$, $(0, 1, 1, 1)$, $(0, 0, 1, 1)$, $(0, 0, 0, 1)$ in $V_4(R)$

Soln:- $V_1 = (1, 1, 1, 1)$, $V_2 = (0, 1, 1, 1)$, $V_3 = (0, 0, 1, 1)$
 $V_4 = (0, 0, 0, 1)$

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$R_2 = R_2 - R_1, \quad R_3 = R_3 - R_1, \quad R_4 = R_4 - R_1$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 = R_3 - R_2$$

$$R_4 = R_4 - R_2$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$S(A) = 4$$

$$\text{No. of unknowns} = 4$$

$$\text{Hence, } S(A) = \text{No. of unknowns}$$

$\therefore$ The given vectors are linearly independent

Linear combination:— $L[S]$

$$\alpha = C_1 \alpha_1 + C_2 \alpha_2 + C_3 \alpha_3$$

If $\alpha = C_1 \alpha_1 + C_2 \alpha_2 + C_3 \alpha_3$ is said to be linear combination when $\alpha_1, \alpha_2, \alpha_3$ are the vectors chosen at a different combination and $C_1, C_2, C_3 \in \mathbb{R}$ together we will get a linear combination which is denoted by " $L[S]$ ".

Basis and dimensions:—
(order of matrix)

* Dimensions: The no. of elements in any basis of V is called $\text{dimension}(V)$ ($\dim(V)$) & it is denoted by " $\dim(V)$ ".

Ex:-i) $[a_{ij}]_{2 \times 2}$ then $\dim(V) = 2 \times 2 = 4$

ii) $\begin{matrix} 1 & \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 1 & 2 \end{bmatrix} \\ 2 & \\ 3 & \end{matrix}_{3 \times 3}$ then $\dim(V) = 3 \times 3 = 9$

* Basis: A subset B of a vector space is called a basis if V consists of linearly independent elements and $V = L[B] \rightarrow$ Linear combination

• Basis = linearly independent + $\dim L[S]$

1) Test whether the set of $S = \{(1, 2, 3), (3, 1, 0), (-2, 1, 3)\}$ is a basis of $V_3(\mathbb{R})$. In case S is not a basis, determine the dimension & the basis of the subspace spanned by S .

Solⁿ:- Consider $|S| = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 0 \\ -2 & 1 & 3 \end{vmatrix}$

$$= 1 \begin{vmatrix} 1 & 0 \\ 1 & 3 \end{vmatrix} - 2 \begin{vmatrix} 3 & 0 \\ -2 & 3 \end{vmatrix} + 3 \begin{vmatrix} 3 & 1 \\ -2 & 1 \end{vmatrix}$$

$$= 1(3-0) - 2(9+0) + 3(3+2)$$

$$= 3 - 18 + 15$$

$$|S| = 0$$

$\therefore S$ is linearly dependent.

Hence, S is not a basis of $V_3(\mathbb{R})$.

$$[S] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 0 \\ -2 & 1 & 3 \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 - 3R_1 \\ R_3 = R_3 + 2R_1 \end{array}$$

$$[S] = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -9 \\ 0 & 5 & 9 \end{bmatrix} \quad \begin{array}{l} R_3 = R_3 + R_2 \\ R_3 = R_3 + R_2 \end{array}$$

$$[S] = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -9 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho(S) = 2$$

Dimension of $V_3(\mathbb{R})$.

$(1, 2, 3), (0, -5, -9)$ are the spanned by S .

• Basis & dimension
 $\Rightarrow$ row interms of row itself.

• Linearly dep & indep
 $\Rightarrow$ row interms of column.

2) Test whether the following set of vectors are basis of $\underset{\substack{\downarrow \\ V_3(R)}}{R^3}$ or not, $S = \{(4, 3, 0), (3, 2, -1), (4, 3, -2)\}$

Solⁿ:- To say its basis it satisfies 2 conditions

i) Linear independent

ii) Linear combination $L[S]$

$$[S] = \begin{vmatrix} 4 & 3 & 0 \\ 3 & 2 & -1 \\ 4 & 3 & -2 \end{vmatrix} \Rightarrow 4 \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} - 3 \begin{vmatrix} 3 & -1 \\ 4 & -2 \end{vmatrix} + 0 \begin{vmatrix} 3 & 2 \\ 4 & 3 \end{vmatrix}$$

$$= 4(-4+3) - 3(-6+4) + 0(9-8)$$

$$= 4(-1) - 3(-2) + 0$$

$$= -4 + 6$$

$$[S] = 2$$

$$2 \neq 0$$

$|S| \neq 0 \therefore S$ is linearly independent

* Hence ~~S is a basis of $R^3 | V_3(R)$~~ *

ii) To check linear combination we should

evaluate generally $a, b, c = x(4, 3, 0) + y(3, 2, -1) + z(4, 3, -2)$

$$a, b, c = (4x, 3x, 0) + (3y, 2y, -y) + (4z, 3z, -2z)$$

$$a, b, c = (4x + 3y + 4z, 3x + 2y + 3z, 0 + y - 2z)$$

$$[A:B] = \begin{bmatrix} 4 & 3 & 4 & : & a \\ 3 & 2 & 3 & : & b \\ 0 & -1 & -2 & : & c \end{bmatrix} \quad \begin{array}{l} R_2 = 4R_2 - 3R_1 \\ 4(2) - 3(3) \\ 8 - 9 \\ -1 \end{array}$$

$$= \begin{bmatrix} 4 & 3 & 4 & : & a \\ 0 & -1 & 0 & : & 4b-3a \\ 0 & -1 & -2 & : & c \end{bmatrix} \quad \begin{array}{l} R_3 = R_3 - R_2 \\ -2 - 0 \\ c - 4b + 3a \end{array}$$

$$= \begin{bmatrix} 4 & 3 & 4 & : & a \\ 0 & -1 & 0 & : & 4b-3a \\ 0 & 0 & -2 & : & c-4b+3a \end{bmatrix}$$

$S(A) = 3 = S(A:B) \rightarrow$ consistency. - system has no solution.
Hence, the given vectors are basis of $\mathbb{R}^3$.

3) Test whether the following set of vectors are basis or not, $S = \{(2, 3, 0), (1, 2, 5), (1, 9, 4)\}$

Solⁿ:- $[S] = \begin{vmatrix} 2 & 3 & 0 \\ 1 & 2 & 5 \\ 1 & 9 & 4 \end{vmatrix} \Rightarrow 2 \begin{vmatrix} 2 & 5 \\ 9 & 4 \end{vmatrix} - 3 \begin{vmatrix} 1 & 5 \\ 1 & 4 \end{vmatrix} + 0 \begin{vmatrix} 1 & 2 \\ 1 & 9 \end{vmatrix}$

$$= 2(8 - 45) - 3(4 - 5) + 0$$

$$= 2(-37) - 3(-1)$$

$$= -74 + 3$$

$$= -71$$

$\therefore S$ is linearly independent.

$$[S] = -71$$

$$-71 \neq 0$$

$$[S] \neq 0$$

ii) To check linear combination

$$a, b, c = x(2, 3, 0) + y(1, 2, 5) + z(1, 9, 4)$$

$$a, b, c = (2x, 3x, 0) + (y, 2y, 5y) + (z, 9z, 4z)$$

$$a, b, c = (2x + y + z, 3x + 2y + 9z, 0 + 5y + 4z)$$

$$a, b, c = (2x + y + z, 3x + 2y + 9z, 0 + 5y + 4z)$$

$$[A:B] = \begin{bmatrix} 2 & 1 & 1 & : & a \\ 3 & 2 & 9 & : & b \\ 0 & 5 & 4 & : & c \end{bmatrix}$$

$$R_2 = R_2 - 3R_1$$

$$= 3 - 3(2)$$

$$= 3 - 6$$

$$= -3$$

$$= \begin{bmatrix} 2 & 1 & 1 & : & a \\ 1 & 1 & 8 & : & b-a \\ 0 & 5 & 4 & : & c \end{bmatrix}$$

$$R_2 = 3R_2 - 2R_3$$

$$= 3(-3) - 2(4)$$

$$= -9 - 8$$

$$= -17$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 15 & 3b-a \\ 0 & 5 & 4 & c \end{array} \right]$$

$$R_2 \leftrightarrow R_1$$

$$[A:B] = \left[\begin{array}{ccc|c} 2 & 1 & 1 & a \\ 0 & 1 & 15 & 3b-a \\ 0 & 5 & 4 & c \end{array} \right] \quad R_2 = 2R_2 - 3R_1$$

$$2(1) - 3(1)$$

$$4 - 3$$

$$18 - 3$$

$$2(b-a) - 3(a)$$

$$2b - 2a - 3a$$

$$R_3 = R_3 - 5R_2$$

$$4 - 5(15)$$

$$4 - 75$$

$$\frac{75 \times 5}{75}$$

$$2b - a$$

$$[A:B] = \left[\begin{array}{ccc|c} 2 & 1 & 1 & a \\ 0 & 1 & 15 & 3b-a \\ 0 & 0 & -71 & c - 15b - 5a \end{array} \right] \quad c - 5(3b-a)$$

Hence, the given vectors forms the basis of $\mathbb{R}^3$.

5) Determine whether the given vectors be in span W if $V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $V_2 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$, $V_3 = \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$, $W = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$

Solⁿ:- $S = \{V_1, V_2, V_3\}$, $W \in L[S]$

$$W = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = xV_1 + yV_2 + zV_3$$

$$\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} + z \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} x + 2y + 2z \\ 0 + 2y + 0 \\ 0 + y + 4z \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 2 & : & 1 \\ 0 & -2 & 0 & : & 3 \\ 0 & 1 & 4 & : & -1 \end{bmatrix} \quad R_3 = 2R_3 + R_2$$

augmented matrix

$$[A:B] = \begin{bmatrix} 1 & 2 & 2 & : & 1 \\ 0 & -2 & 0 & : & 3 \\ 0 & 0 & 8 & : & 1 \end{bmatrix} \quad \begin{aligned} \rho(A) &= 3 \\ \rho(A:B) &= 3 \end{aligned}$$

$$\therefore \rho(A) = 3 = \rho(A:B)$$

consistency - solution exists.

Hence, ~~W is a sub~~ $W \in L[S]$.

6) For what value of h will Y be in span $\{V_1, V_2, V_3\}$. If $V_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$, $V_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}$ & $V_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$ and $Y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$.

Solⁿ:- $Y \in L[S]$

$$\begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix} = xV_1 + yV_2 + zV_3$$

$$\begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix} = x \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix} + z \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix} = \begin{bmatrix} x+5y-3z \\ -x-4y+z \\ -2x-7y+0 \end{bmatrix}$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array} \right] \quad \begin{array}{l} R_2 = R_2 + R_1 \\ R_3 = \end{array} \quad \begin{array}{l} -4+5 \\ -2-2 \\ 1-3 \end{array}$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & -4 & -2 & -1 \\ -2 & -7 & 0 & h \end{array} \right] \quad \begin{array}{l} R_3 = R_3 + 2R_1 \\ = -2 - 2(1) \\ = -7 + 2(5) \\ = -7 + 10 \\ = h + 2(-4) \\ = h - 8 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & -4 & -2 & -1 \\ 0 & -6 & 0 & h-8 \end{array} \right] \quad R_3 = R_3 - 3R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & -4 & -2 & -1 \\ 0 & 0 & 0 & h-5 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & -4 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Here $h-5=0 \Rightarrow \boxed{h=5}$

$\left. \begin{array}{l} \rho(A) = 2, \rho[A:B] = 3 \\ \rho(A) \neq \rho[A:B] \end{array} \right\} \therefore \text{Inconsistency - sol}^n \text{ does not exist}$

$\rho(A) = 2, \rho[A:B] = 3$
 consistency
 non exist
 $\therefore \underline{w \notin L(A)}$

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7) Express the vector $(3, 5, 2)$ as a linear combination of the vectors $(1, 1, 0)$, $(2, 3, 0)$, $(0, 0, 1)$ of $V_3(R)$.

Solⁿ:- $(3, 5, 2) = c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3$

$$(3, 5, 2) = c_1(1, 1, 0) + c_2(2, 3, 0) + c_3(0, 0, 1)$$

$$(3, 5, 2) = (c_1, c_1, 0) + (2c_2, 3c_2, 0) + (0, 0, c_3)$$

$$(3, 5, 2) = (c_1 + 2c_2, c_1 + 3c_2, c_3)$$

$$\Rightarrow \boxed{c_3 = 2}, \quad c_1 + 2c_2 = 3, \quad c_1 + 3c_2 = 5$$

$$\begin{array}{r} c_1 + 2c_2 = 3 \\ (-) \quad (-) \quad (-) \\ \hline c_2 = 2 \end{array}$$

$$c_1 + 3(2) = 5$$

$$c_1 + 6 = 5$$

$$c_1 = 5 - 6$$

$$\boxed{c_1 = -1}$$

$$\left\{ \begin{array}{l} -1 + 4 = 3 \\ 3 = 3 \\ \text{LHS} = \text{RHS} \end{array} \right\}$$

8) Express the vector $(2, -5, 3)$ as a linear combination of the vectors $(1, -3, 2)$, $(2, -4, -1)$, $(1, -5, 7)$ of $V_3(R)$.

Solⁿ:- $(2, -5, 3) = c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3$

$$(2, -5, 3) = c_1(1, -3, 2) + c_2(2, -4, -1) + c_3(1, -5, 7)$$

$$(2, -5, 3) = (c_1, -3c_1, 2c_1) + (2c_2, -4c_2, -c_2) +$$

$$(c_3, -5c_3, 7c_3)$$

$$(2, -5, 3) = (c_1 + 2c_2 + c_3, -3c_1 - 4c_2 - 5c_3, 2c_1 - c_2 + 7c_3)$$

$$(2c_1 - c_2 + 7c_3)$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ -3 & -4 & -5 & -5 \\ 2 & -1 & 7 & 3 \end{array} \right]$$

$$R_2 = R_2 + 3R_1, \quad R_3 = R_3 - 2R_1$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 3 & : & 2 \\ 0 & 2 & -2 & : & 1 \\ 0 & -5 & 5 & : & -1 \end{bmatrix}$$

$$\begin{aligned} 2 - 2(1) &= -5 + 3(2) = -5 + 6 = 1 \\ -1 - 2(2) &= -3 + 3(1) = 0 \\ -1 - 4 &= -4 + 3(2) = -4 + 6 = 2 \\ 7 - 2(3) &= -5 + 3(3) = -5 + 9 = 4 \\ 7 - 6 &= 1 \\ 3 - 2(2) &= 3 - 4 = -1 \end{aligned}$$

$$R_3 = 2R_3 + 5R_2$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 3 & : & 2 \\ 0 & 2 & -2 & : & 1 \\ 0 & 0 & 0 & : & 3 \end{bmatrix}$$

$$\begin{matrix} 2(5) + 5(-2) \\ 10 - 10 \end{matrix} \quad \rho(A) = 2, \quad \rho(A:B) = 3$$

$$\text{Here, } \rho(A) \neq \rho(A:B)$$

Hence, the given vectors ~~can~~ not ~~linear~~ express as a linear combination.

① Identify the k value for which the vectors $(1, -2, k)$, $(2, -1, 5)$ and $(3, -5, 7k)$ are linearly dependent.

$$\underline{\text{Sol}^n}:- A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ k & 5 & 7k \end{bmatrix}$$

$$\underline{\text{Sol}^n}:- \begin{vmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ k & 5 & 7k \end{vmatrix} = 1 \begin{vmatrix} -1 & -5 \\ 5 & 7k \end{vmatrix} - 2 \begin{vmatrix} -2 & -5 \\ k & 7k \end{vmatrix} + 3 \begin{vmatrix} -2 & -1 \\ k & 5 \end{vmatrix}$$

$$0 = 1(-7k + 25) - 2(-14k + 5k) + 3(-10 + k)$$

$$0 = -7k + 25 + 28k - 10k - 30 + 3k$$

$$0 = 14k - 5$$

$$14k = 5$$

$$\boxed{k = 5/14}$$

$(-2, 0, 1), (3, 2, 5), (6, -1, 1), (7, 0, -2)$ vectors are linearly dependent or independent?

Solⁿ:- $\begin{bmatrix} -2 & 3 & 6 & 7 \\ 0 & 2 & -1 & 0 \\ 1 & 5 & 1 & -2 \end{bmatrix} \quad R_1 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & 5 & 1 & -2 \\ 0 & 2 & -1 & 0 \\ -2 & 3 & 6 & 7 \end{bmatrix}$$

$$R_3 = R_3 + 2R_1$$

$$3 + 2(5) = 3 + 10$$

$$6 + 2(1) = 6 + 2$$

$$7 + 2(-2) = 7 - 4 = 3$$

$$\begin{bmatrix} 1 & 5 & 1 & -2 \\ 0 & 2 & -1 & 0 \\ 0 & 13 & 8 & 3 \end{bmatrix}$$

$$R_3 = 2R_3 - 13R_2 \rightarrow \begin{bmatrix} 1 & 5 & 1 & -2 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 29 & 6 \end{bmatrix}$$

$$= 2(8) - 13(-1)$$

$$= 16 + 13$$

$$= 2(3) - 13(0)$$

Here, $\rho(A) = 3$, NO of unknowns = 3

$\rho(A) \neq$ NO. of unknowns

Hence, the given vectors are linearly independent.

MQP:- (Module - 01)
Show that set of all 2×2 matrices with real numbers forms a vector space over the field of real numbers.

Solⁿ:- We know that to prove a vector space we need to satisfy the following conditions

i) Vector addition

ii) Scalar multiplication

Let us consider in general 2×2 matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ \& } B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$$

Next, ① $A+B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} a+x & b+y \\ c+z & d+w \end{bmatrix}_{2 \times 2}$

vector addition

② Scalar multiplication

$$\alpha A = \alpha \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{bmatrix}_{2 \times 2}$$

$\therefore$ All the 2×2 matrices forms a vector space with the real numbers.

1) Express the vector $(3, -7, 6)$ as a linear combination of the vectors $(1, -3, 2)$ $(2, 4, 1)$ $(1, 1, 1)$ of $\mathbb{R}^3$
 α_1 α_2 α_3

Solⁿ:- $(3, -7, 6) = C_1 \alpha_1 + C_2 \alpha_2 + C_3 \alpha_3$

$$(3, -7, 6) = C_1(1, -3, 2) + C_2(2, 4, 1) + C_3(1, 1, 1)$$

$$(3, -7, 6) = (C_1, -3C_1, 2C_1) + (2C_2, 4C_2, C_2) + (C_3, C_3, C_3)$$

$$(3, -7, 6) = (C_1 + 2C_2 + C_3, -3C_1 + 4C_2 + C_3, 2C_1 + C_2 + C_3)$$

$$C_1 + 2C_2 + C_3 = 3$$

$$-3C_1 + 4C_2 + C_3 = -7$$

$$2C_1 + C_2 + C_3 = 6$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ -3 & 4 & 1 & : & -7 \\ 2 & 1 & 1 & : & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 3R_1 \quad -3+ \\ R_3 \rightarrow R_3 - 2R_1$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ 0 & 10 & 4 & : & 2 \\ 0 & -3 & -1 & : & 0 \end{bmatrix}$$

$$R_3 \Rightarrow 10R_3 + 3R_2$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ 0 & 10 & 4 & : & 2 \\ 0 & 0 & 2 & : & 6 \end{bmatrix}$$

$$\rho(A) = 3$$

$$\rho[A:B] = 3$$

$$\therefore \rho(A) = \rho[A:B] \\ 3 = 3$$

Hence, the given vectors express as a linear combination.

24) Identify whether the following set of vectors are basis of R^3 or not

$$S = \{(1, 0, 0), (2, 2, 0), (4, 3, -2)\}$$

Soln:- To say it's basis it satisfies 2 conditions

i) Linear independent

ii) Linear combination

$$i) [S] = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 4 & 3 & -2 \end{vmatrix} \Rightarrow 1 \begin{vmatrix} 2 & 0 \\ 3 & -2 \end{vmatrix} - 0 \begin{vmatrix} 2 & 0 \\ 4 & -2 \end{vmatrix} + 0 \begin{vmatrix} 2 & 2 \\ 4 & 3 \end{vmatrix}$$

$$= 1(4 - 0) - 0(-4 - 4) + 0(6 - 8)$$

$$= 1(4) - 0(-8) + 0(-2)$$

$$[S] = -4$$

$$-4 \neq 0$$

$$|S| \neq 0$$

$\therefore S$ is linearly independent.

ii) To check linear combination we should equate generally $a.b.c = x(1.0.0) + y(2.2.0) + z(4.3.2)$

$$a.b.c = (x + (2y + 4z), 2y + 3z, -2z)$$

$$a.b.c = (x + 2y + 4z, 2y + 3z, -2z)$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 4 & a \\ 0 & 2 & 3 & b \\ 0 & 0 & -2 & c \end{array} \right] \quad R_2 \rightarrow \frac{1}{2}R_2$$

$$\rho(A) = 3, \rho(A:B) = 3$$

$$\therefore \rho(A) = \rho(A:B)$$

Hence, the given vectors are basis of R^3

25) Identify whether the following set of vectors are basis of R^3 or not.

$$S = \{(1, -2, 3), (5, 6, -1), (3, 2, 1)\}.$$

Soln:- To say if ~~its~~ basis it satisfies ② condition

i) linear independent

ii) linear combination

$$\begin{aligned} |S| &= \begin{vmatrix} 1 & -2 & 3 \\ 5 & 6 & -1 \\ 3 & 2 & 1 \end{vmatrix} \Rightarrow 1 \begin{vmatrix} 6 & -1 \\ 2 & 1 \end{vmatrix} + 2 \begin{vmatrix} 5 & -1 \\ 3 & 1 \end{vmatrix} + 3 \begin{vmatrix} 5 & 6 \\ 3 & 2 \end{vmatrix} \\ &= 1(6+2) + 2(5+3) + 3(10-18) \\ &= 1(8) + 2(8) + 3(-8) \\ &= 8 + 16 - 24 \end{aligned}$$

$$|S| = 0$$

~~linear independent~~

~~linear combination~~

$\therefore S$ is linearly ~~in~~dependent.

ii) To check linear combination we should equate generally $a.b.c = x(1, -2, 3) + y(5, 6, -1) + z(3, 2, 1)$

$$a.b.c = (x, -2x, 3x) + (5y, 6y, -4) + (3z, 2z, z)$$

$$a.b.c = (x+5y+3z, -2x+6y+2z, 3x-y+z)$$

$$[A:B] = \begin{bmatrix} 1 & 5 & 3 & : & a \\ -2 & 6 & 2 & : & b \\ 3 & -1 & 1 & : & c \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad \begin{array}{l} -2-2(-) \\ 6+2(1) \end{array}$$

$$[A:B] = \begin{bmatrix} 1 & 5 & 3 & : & a \\ 0 & 16 & 8 & : & b+2a \\ 0 & -16 & -8 & : & c-3a \end{bmatrix} \quad \begin{array}{l} 1-3(3) \\ 1-9 \end{array} \quad \begin{array}{l} 2+2(3) \\ 2+6 \end{array}$$

$$R_3 \rightarrow R_3 + R_2$$

$$[A:B] = \begin{bmatrix} 1 & 5 & 3 & : & a \\ 0 & 16 & 8 & : & b+2a \\ 0 & 0 & 0 & : & (c-3a) + (b+2a) \end{bmatrix}$$

$$\rho(A) = 2, \quad \rho(A:B) = 3$$

$$\rho(A) \neq \rho(A:B)$$

Hence, the given vectors are not basis of $\mathbb{R}^3$.

26) Test whether the set $\{(1, 2, 1), (3, 4, -7), (3, 1, 5)\}$ is a basis of $V_3(\mathbb{R})$ or not.

Solⁿ:- To say it forms basis it satisfies ② condⁿ

i) Linear independent

ii) Linear combination

$$|S| = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & -7 \\ 3 & 1 & 5 \end{vmatrix} \Rightarrow 1 \begin{vmatrix} 4 & -7 \\ 1 & 5 \end{vmatrix} - 2 \begin{vmatrix} 3 & -7 \\ 3 & 5 \end{vmatrix} + 1 \begin{vmatrix} 3 & 4 \\ 3 & 1 \end{vmatrix}$$

$$= 1(20+7) - 2(15+21) + 1(3-12)$$

$$= 1(27) - 2(36) + 1(-9)$$

$$= 27 - 72 - 9$$

$$|S| = 27 - 81$$

$$|S| = -54$$

$\therefore S$ is linearly independent.

i) To check linear combination we should evaluate generally $a.b.c = x(1, 2, 1) + y(3, 4, -7) + z(3, 1, 5)$

$$a.b.c = (x, 2x, x) + (3y, 4y, -7y) + (3z, z, 5z)$$

$$a.b.c = (x+3y+3z), (2x+4y+z), (x-7y+5z)$$

$$[A:B] = \begin{bmatrix} 1 & 3 & 3 & : & a \\ 2 & 4 & 1 & : & b \\ 1 & -7 & 5 & : & c \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \quad \begin{array}{l} 2-2(1) \\ 4-2(3) \\ 1-1 \end{array}$$

$$[A:B] = \begin{bmatrix} 1 & 3 & 3 & : & a \\ 0 & -2 & -5 & : & b-2a \\ 0 & -10 & 2 & : & c-a \end{bmatrix} \quad \begin{array}{l} 1-2(3) \\ 1-1 \end{array} \quad \begin{array}{l} 4-2(3) \\ 4- \\ 1-3(3) \\ -1-3 \end{array}$$

$$R_3 \rightarrow R_3 - 5R_2$$

$$[A:B] = \begin{bmatrix} 1 & 3 & 3 & : & a \\ 0 & -2 & -5 & : & b-2a \\ 0 & 0 & 27 & : & 9a-5b+c \end{bmatrix} \quad \begin{array}{l} 2-5(-5) \\ -10-5(-2) \\ -10+10 \end{array} \quad \begin{array}{l} 2+25 \\ -10-5(-2) \\ -10+10 \end{array}$$

$$\text{Here, } \rho(A) = 3, \rho(A:B) = 3$$

$$\rho(A) = \rho(A:B)$$

Hence, the given vectors are basis of R^3 .

2) Test whether the set $B = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis of $V_3(R)$.

Solⁿ:- To form a basis it satisfies 2 condⁿ.

i) Linear independent

ii) Linear combination

$$|B| = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} \Rightarrow 1 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= 1(0-1) - 1(1-0) + 0(1-0)$$

$$= 1(-1) - 1(1) + 0$$

$$|B| = -2$$

$|B| \neq 0 \therefore S$ is a linear independent
 $-2 \neq 0$

ii) To check linear combination we should equate
 generally $a, b, c = x(1, 1, 0) + y(1, 0, 1) + z(0, 1, 1)$

$$a, b, c = (x, x, 0) + (y, 0, y) + (0, z, z)$$

$$a, b, c = (x+y+0), (x+0+z), (0+y+z)$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 1 & 0 & 1 & b \\ 0 & 1 & 1 & c \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 0 & -1 & 1 & b-a \\ 0 & 0 & 1 & c-a \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 + R_2 \end{array}$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 0 & -1 & 1 & b-a \\ 0 & 1 & 1 & c-a \end{array} \right] \quad [A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & a \\ 0 & -1 & 1 & b-a \\ 0 & 0 & 2 & -a+b+c \end{array} \right]$$

$$\text{Here, } \rho(A) = 3, \rho(A:B) = 3$$

$$\rho(A) = \rho(A:B)$$

Hence, the given vectors are basis of R^3 .

28) Determine the dimension and the basis of
 the subspace spanned by the vectors $\{(2, 4, 2),$
 $(1, -1, 0), (1, 2, 1), (0, 3, 1)\}$

$$\underline{\text{Soln}}:- A = \left[\begin{array}{ccc} 2 & 4 & 2 \\ 1 & -1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$A = \left[\begin{array}{ccc} 1 & -1 & 0 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 6 & 2 \\ 0 & 3 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{2} R_2$$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 1 \\ 0 & 3 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1, R_4 \rightarrow R_4 - R_2$$

$$\text{Here, } \rho(A) = 2$$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Here we take non-zero elements from the reduced matrix Basis = $\{(1, -1, 0), (0, 3, 1)\}$

∴ The dimension is the number of vectors in the basis. Dimension = 2 $\rightarrow \boxed{\rho(A) = \dim}$

1) Identify whether the following set of vectors are linearly independent $\{(2, 2, 1), (-4, 6, 5), (1, 0, 0)\}$

Solⁿ: Given: $V_1 = (2, 2, 1), V_2 = (-4, 6, 5), V_3 = (1, 0, 0)$

$$A = \begin{bmatrix} 2 & -4 & 1 \\ 2 & 6 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$R_1 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 6 & 0 \\ 2 & -4 & 1 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -4 & 0 \\ 0 & -14 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{14}{4} R_2$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Here, } \rho(A) = 3$$

∴ $\rho(A) = \text{NO. of unknowns}$

$$\rho(A) = 3$$

Hence, the given set of vectors are linearly independent

16) show that $W = (-9, 7, 4, 8)$ is in the subspace of $\mathbb{R}^4$ spanned by $(7, -4, 2, 9)$, $(-4, 5, -1, -7)$, $(-9, 4, 4, -7)$

sol:- Let $V_1 = (7, -4, 2, 9)$

$$V_2 = (-4, 5, -1, -7)$$

$$V_3 = (-9, 4, 4, -7)$$

$$(-9, 7, 4, 8) = C_1 V_1 + C_2 V_2 + C_3 V_3$$

$$(-9, 7, 4, 8) = C_1 (7, -4, 2, 9) + C_2 (-4, 5, -1, -7) + C_3 (-9, 4, 4, -7)$$

$$(-9, 7, 4, 8) = (7C_1, -4C_1, 2C_1, 9C_1) + (-4C_2, 5C_2, -C_2, -7C_2) + (-9C_3, 4C_3, 4C_3, -7C_3)$$

$$(-9, 7, 4, 8) = (7C_1 - 4C_2 - 9C_3, -4C_1 + 5C_2 + 4C_3, 2C_1 - C_2 + 4C_3, 9C_1 - 7C_2 - 7C_3)$$

$$7C_1 - 4C_2 - 9C_3 = -9$$

$$-4C_1 + 5C_2 + 4C_3 = 7$$

$$2C_1 - C_2 + 4C_3 = 4$$

$$9C_1 - 7C_2 - 7C_3 = 8$$

$$[A:W] = \begin{bmatrix} 7 & -4 & -9 & : & 9 \\ -4 & 5 & 4 & : & 7 \\ 2 & -1 & 4 & : & 4 \\ -9 & -7 & -7 & : & 8 \end{bmatrix}$$

$$[A:W] = \begin{bmatrix} 2 & -1 & 4 & : & 4 \\ -4 & 5 & 4 & : & 7 \\ 7 & -4 & -9 & : & -9 \\ -9 & -7 & -7 & : & 8 \end{bmatrix} \quad \begin{array}{l} R_1 \leftrightarrow R_3 \\ R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - \frac{7}{2}R_1 \\ R_4 \rightarrow R_4 - \frac{9}{2}R_1 \end{array}$$

$$[A:W] = \begin{bmatrix} 2 & -1 & 4 & : & 4 \\ 0 & 3 & 12 & : & 15 \\ 0 & -\frac{1}{2} & -23 & : & -23 \\ 0 & -\frac{5}{2} & -25 & : & -10 \end{bmatrix} \quad \begin{array}{l} R_3 = R_3 + \frac{1}{2}R_2 \\ R_4 = R_4 + \frac{5}{2}R_2 \end{array}$$

$$[A:W] = \begin{bmatrix} 2 & -1 & 4 & : & 4 \\ 0 & 1 & 4 & : & 5 \\ 0 & 0 & -21 & : & -\frac{9}{2} \\ 0 & 0 & -15 & : & \frac{5}{2} \end{bmatrix} \quad \begin{array}{l} R_4 \rightarrow R_4 - \frac{5}{7}R_3 \\ [A:W] = \begin{bmatrix} 2 & -1 & 4 & : & 4 \\ 0 & 1 & 4 & : & 5 \\ 0 & 0 & -21 & : & -\frac{9}{2} \\ 0 & 0 & 0 & : & \frac{120}{7} \end{bmatrix} \end{array}$$

Here, $0 \neq \frac{120}{7}$

Hence, No the vector $W = (-9, 7, 4, 8)$ is not in the subspace spanned by the given vectors because the system of linear equations has no solⁿ.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1$$

$$R_2 \rightarrow R_2$$

$$R_3 \rightarrow R_3$$

$$R_4 \rightarrow R_4$$

$$R_5 \rightarrow R_5$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

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